

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- AI
PRCATICAL NO: 2

AIM: A* Search and Recursive Best-First Search

- Implement the A* Search algorithm for solving a pathfinding problem.
- Implement the Recursive Best-First Search algorithm for the same problem.
- Compare the performance and effectiveness of both algorithms.

```
# Iterative Deepening Depth First Search (IDDFS)
# Mumbai to Pune Route Search

graph = {
    "Mumbai": ["Eastern Express Highway", "Atal Setu", "Dadar Railway
Station"],

    # Route 1 (Car)
    "Eastern Express Highway": ["Sion"],
    "Sion": ["Panvel"],
    "Panvel": ["Lonavala"],
    "Lonavala": ["Pimpri Chinchwad"],
    "Pimpri Chinchwad": ["Pune"],

    # Route 2 (Bike)
    "Atal Setu": ["JNPT"],
    "JNPT": ["Khalapur"],
    "Khalapur": ["Lonavala"],

    # Route 3 (Train)
    "Dadar Railway Station": ["Karjat"],
    "Karjat": ["Lonavala Railway Station"],
    "Lonavala Railway Station": ["Shivajinagar"],
    "Shivajinagar": ["Pune"],

    "Pune": []
}

# Depth Limited Search
def DLS(node, goal, limit, path, visited):
```

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SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- AI
PRCATICAL NO: 2

```
print(node, end=" ")

if node == goal:
    path.append(node)
    return True

if limit <= 0:
    return False

visited.add(node)

for neighbour in graph[node]:
    if neighbour not in visited:
        if DLS(neighbour, goal, limit - 1, path, visited):
            path.insert(0, node)
            return True

return False

# Iterative Deepening DFS
def IDDFS(start, goal, max_depth):

    for depth in range(max_depth + 1):

        print("\nSearching at Limit:", depth)

        path = []
        visited = set()

        found = DLS(start, goal, depth, path, visited)

        if found:
            print("\n\nFound")
            print("IDDFS Traversal from", start, "to", goal, "is:")
            print(" -> ".join(path))
            return
```

NAME: TANISHKA KUTWAL
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SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- AI
PRCATICAL NO: 2

```
        else:
            print("\n\nNot Found!")

        print("-" * 50)

source = "Mumbai"
destination = "Pune"

IDDFS(source, destination, 6)

print("\n")
print("=" * 50)
print("Available Routes")
print("=" * 50)

print("\nRoute 1 (Car)")
print("Mumbai -> Eastern Express Highway -> Sion -> Panvel -> Lonavala -> Pimpri Chinchwad -> Pune")

print("\nRoute 2 (Bike)")
print("Mumbai -> Atal Setu -> JNPT -> Khalapur -> Lonavala -> Pimpri Chinchwad -> Pune")

print("\nRoute 3 (Train)")
print("Mumbai -> Dadar Railway Station -> Karjat -> Lonavala Railway Station -> Shivajinagar -> Pune")
```

OUTPUT:

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT :- AI

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```
PS C:\Users\mvlul\Documents\YV889CS\AI> cd 'C:\Users\mvlul\Documents\YV889CS\AI'; & 'C:\Users\mvlul\AppData\Local\Programs\Python\Python314\python.exe' 'C:\Users\mvlul\.vscode\extensions\ms-python.debugpy-2026.6.0-wn32-x64\bundle\libs\debugpy\launcher' '63665' '-.' 'C:\Users\mvlul\Documents\YV889CS\AI\prac 2'
```

Mumbai

Not Found!

Searching at Limit: 1

Mumbai Eastern Express Highway Atal Setu Dadar Railway Station

Not Found!

Searching at Limit: 2

Mumbai Eastern Express Highway Sion Atal Setu JNPT Dadar Railway Station Karjat

Not Found!

Searching at Limit: 3

Mumbai Eastern Express Highway Sion Panvel Atal Setu JNPT Khalapur Dadar Railway Station Karjat Lonavala Railway Station

Not Found!

Searching at Limit: 4

Mumbai Eastern Express Highway Sion Panvel Lonavala Atal Setu JNPT Khalapur Lonavala Dadar Railway Station Karjat Lonavala Railway Station Shivajinagar

Not Found!

Searching at Limit: 5

Mumbai Eastern Express Highway Sion Panvel Lonavala Pinpri Chinchwad Atal Setu JNPT Khalapur Dadar Railway Station Karjat Lonavala Railway Station Shivajinagar Pune

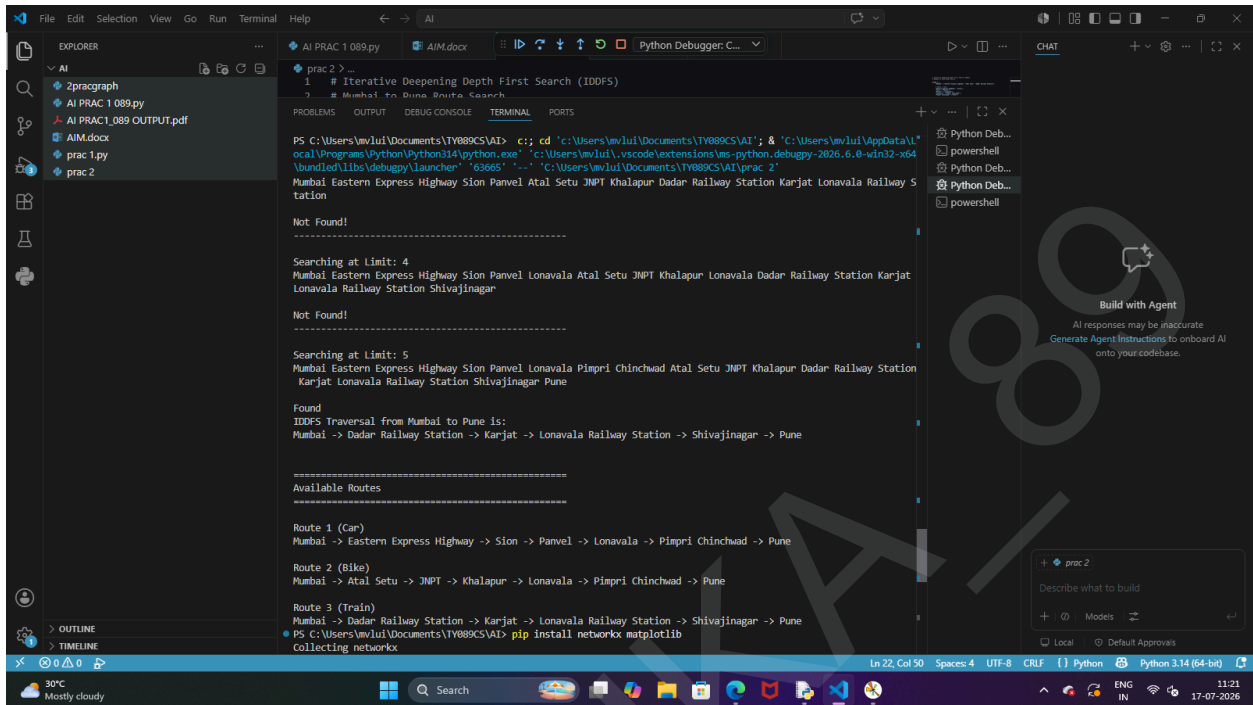
Found

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE

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GRAPH CODE:

```
import networkx as nx
import matplotlib.pyplot as plt

# Create graph
G = nx.Graph()

# Nodes with heuristic values
heuristics = {
    "Mumbai\nh(n)=20": 20,
    "Eastern Express\nHighway\nh(n)=16": 16,
    "Atal Setu\nh(n)=14": 14,
    "Dadar Railway\nStation\nh(n)=15": 15,
    "Panvel\nh(n)=11": 11,
    "Khalapur\nh(n)=10": 10,
    "Karjat\nh(n)=9": 9,
    "Lonavala\nh(n)=6": 6,
    "Lonavala Railway\nh(n)=5": 5,
    "Pimpri-Chinchwad\nh(n)=2": 2,
    "Shivajinagar\nh(n)=2": 2,
    "Pune\nh(n)=0": 0
}
```

NAME: TANISHKA KUTWAL
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SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- AI
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```
}

# Add nodes
for node in heuristics:
    G.add_node(node)

# Add edges (distance in km)
edges = [

    # Route 1 (Car)
    ("Mumbai\nh(n)=20", "Eastern Express\nHighway\nh(n)=16", 5),
    ("Eastern Express\nHighway\nh(n)=16", "Panvel\nh(n)=11", 12),
    ("Panvel\nh(n)=11", "Lonavala\nh(n)=6", 18),
    ("Lonavala\nh(n)=6", "Pimpri-Chinchwad\nh(n)=2", 12),
    ("Pimpri-Chinchwad\nh(n)=2", "Pune\nh(n)=0", 8),

    # Route 2 (Bike)
    ("Mumbai\nh(n)=20", "Atal Setu\nh(n)=14", 8),
    ("Atal Setu\nh(n)=14", "Khalapur\nh(n)=10", 10),
    ("Khalapur\nh(n)=10", "Lonavala\nh(n)=6", 15),

    # Route 3 (Train)
    ("Mumbai\nh(n)=20", "Dadar Railway\nStation\nh(n)=15", 6),
    ("Dadar Railway\nStation\nh(n)=15", "Karjat\nh(n)=9", 11),
    ("Karjat\nh(n)=9", "Lonavala Railway\nh(n)=5", 14),
    ("Lonavala Railway\nh(n)=5", "Shivajinagar\nh(n)=2", 10),
    ("Shivajinagar\nh(n)=2", "Pune\nh(n)=0", 6),

    # Connecting edges
    ("Lonavala\nh(n)=6", "Lonavala Railway\nh(n)=5", 5),
    ("Pimpri-Chinchwad\nh(n)=2", "Shivajinagar\nh(n)=2", 7),
]

G.add_weighted_edges_from(edges)

# Node positions
pos = {
    "Mumbai\nh(n)=20": (0, 4),
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- AI
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```
"Eastern Express\nHighway\nh(n)=16": (-3, 2),
"Atal Setu\nh(n)=14": (0, 2),
"Dadar Railway\nStation\nh(n)=15": (3, 2),

"Panvel\nh(n)=11": (-3, 0),
"Khalapur\nh(n)=10": (0, 0),
"Karjat\nh(n)=9": (3, 0),

"Lonavala\nh(n)=6": (-1.5, -2),
"Lonavala Railway\nh(n)=5": (1.5, -2),

"Pimpri-Chinchwad\nh(n)=2": (-1.5, -4),
"Shivajinagar\nh(n)=2": (1.5, -4),

"Pune\nh(n)=0": (0, -6),
}

# Draw nodes
nx.draw_networkx_nodes(
    G,
    pos,
    node_color="lightblue",
    node_size=2200,
    edgecolors="black"
)

# Draw labels
nx.draw_networkx_labels(
    G,
    pos,
    font_size=8,
    font_weight="bold"
)

# Draw all edges
nx.draw_networkx_edges(
    G,
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- AI
PRCATICAL NO: 2

```
pos,
edge_color="gray",
width=2
)

# Highlight Car Route (Optimal)
car_route = [
    ("Mumbai\nh(n)=20", "Eastern Express\nHighway\nh(n)=16"),
    ("Eastern Express\nHighway\nh(n)=16", "Panvel\nh(n)=11"),
    ("Panvel\nh(n)=11", "Lonavala\nh(n)=6"),
    ("Lonavala\nh(n)=6", "Pimpri-Chinchwad\nh(n)=2"),
    ("Pimpri-Chinchwad\nh(n)=2", "Pune\nh(n)=0"),
]

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=car_route,
    edge_color="orange",
    width=4
)

# Edge labels
edge_labels = {(u, v): f"{w} km" for u, v, w in edges}

nx.draw_networkx_edge_labels(
    G,
    pos,
    edge_labels=edge_labels,
    font_color="red",
    font_size=8
)

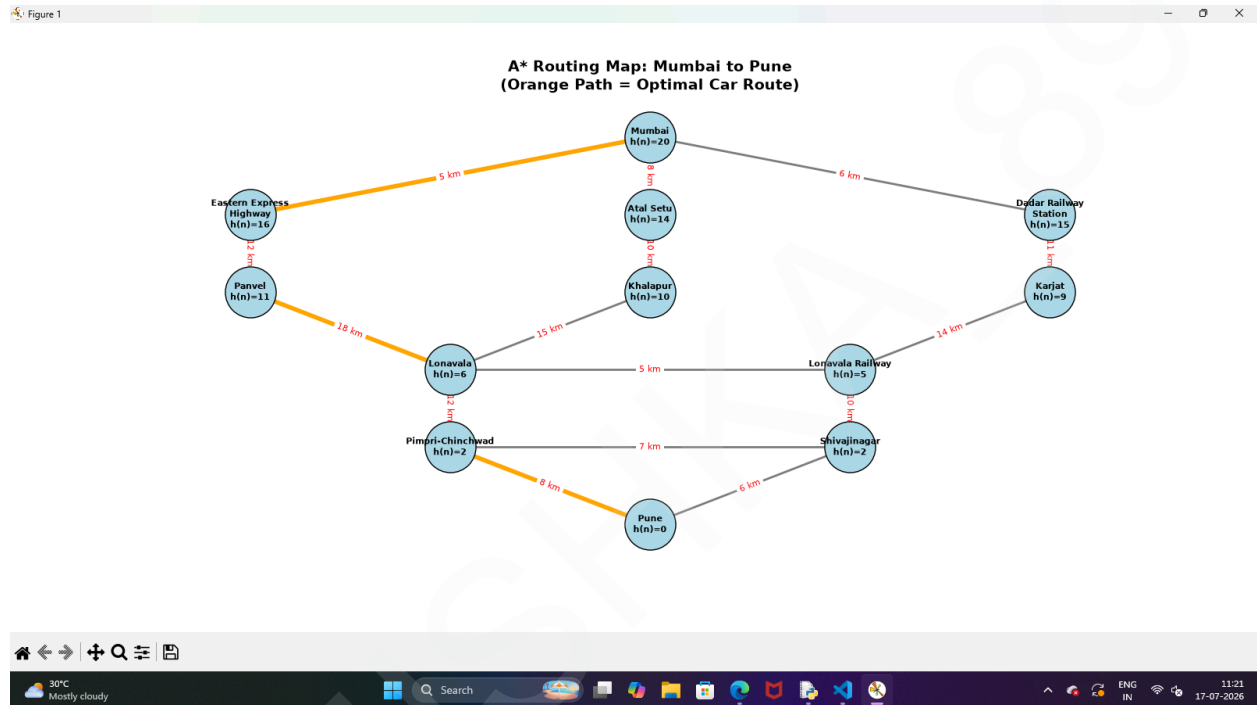
plt.title(
    "A* Routing Map: Mumbai to Pune\n"
    "(Orange Path = Optimal Car Route)",
    fontsize=14,
    fontweight="bold"
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
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```
)  
  
plt.axis("off")  
plt.show()
```

OUTPUT:



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